

A class of locally indicable LOT groups

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Abstract

This paper is concerned with the problem of asphericity of 2-complexes associated to labeled oriented trees (LOTs), which are combinatorial models of ribbon disc complements. The concept of good enumeration for LOTs is introduced and applied to the study of asphericity and local indicability. We describe a large class of LOTs, which includes some families of non-injective LOTs of any diameter and complexity, that present locally indicable groups.

Keywords Labeled oriented trees, group presentations, 2-complexes, asphericity, local indicability.

Mathematics Subject Classification 57M20, 57M35, 06A07, 20F05

1 Introduction

A ribbon n -disc ($n \geq 2$) is a proper embedding $k : D^n \hookrightarrow D^{n+2}$, such that $r \circ k : D^n \rightarrow \mathbb{R}$ is a Morse function with no critical points of index more than 1, where $r : D^{n+2} \rightarrow \mathbb{R}$ is the function $x \mapsto \|x\|$. The homotopy of ribbon disc complements is similar to that of knot complements, and they are therefore conjectured to be aspherical as well. The asphericity of ribbon disc complements is an important special case of the Whitehead Conjecture, which states that subcomplexes of aspherical 2-complexes are aspherical.

Definition 1.1. A *labeled oriented graph (LOG)* $\Gamma = (E, V, s, t, \lambda)$ consists of two sets E, V of edges and vertices, and three maps $s, t, \lambda : E \rightarrow V$ called respectively source, target and label.

Γ is said to be a *labeled oriented tree (LOT)* when the underlying graph is a tree.

Associated to a LOT Γ there is a *LOT presentation* $P(\Gamma)$ with generating set in correspondence to the vertices of Γ and a relator associated to every edge in the following way. For an edge e with source s_e , target t_e and label λ_e , there is a relation $s_e \lambda_e = \lambda_e t_e$, that is, a relator $\lambda_e t_e \lambda_e^{-1} s_e^{-1}$. The group presented by $P(\Gamma)$ is called the LOT group and denoted $G(\Gamma)$. For example, the presentation

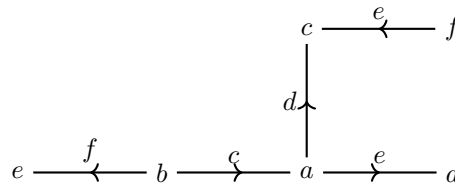


Figure 1: A labeled oriented tree.

associated to the LOT of Figure 1 is

$$P = \langle a, b, c, d, e, f \mid ece^{-1}f^{-1}, dcd^{-1}a^{-1}, ede^{-1}a^{-1}, cac^{-1}b^{-1}, fef^{-1}b^{-1} \rangle$$

Note that the deficiency of a LOT presentation is always 1 since the underlying graph of a LOT is a tree.

Given a LOT Γ , a *sub-LOT* of Γ is a subgraph Δ such that Δ is a tree and $\lambda(e) \in \Delta$ for every edge $e \in \Delta$.

The LOT *complex* K_Γ associated to a given LOT Γ is the standard 2-complex of the LOT presentation. Concretely, K_Γ has a unique 0-cell, one 1-cell for each vertex of Γ , and one 2-cell for each edge of Γ . The attaching map for the 2-cell associated to an edge e spells the relator $\lambda_e t_e \lambda_e^{-1} s_e^{-1}$. The fundamental group of K_Γ is isomorphic to $G(\Gamma)$. A LOT Γ is called *aspherical* if the complex K_Γ is aspherical.

Howie showed in [How85] that LOT complexes are 2-dimensional spines of ribbon disc complements. Concretely, he associated a LOT Γ to every ribbon disc in such a way that K_Γ has the (simple) homotopy type of the ribbon disc complement. Moreover, every LOT corresponds to some ribbon disc, therefore the asphericity of ribbon disc complements is equivalent to the asphericity of LOT complexes.

Howie had proved in [How83] that if a 2-complex L can be 3-deformed to a point, and e is a 2-cell of L , then the subcomplex $K = L - e$ can be 3-deformed to a LOT complex. A consequence of this fact is that if the Andrews-Curtis conjecture were true, then the asphericity of LOT complexes would imply the compact case of the Whitehead conjecture. Recall that the Andrews-Curtis conjecture asserts that every contractible 2-complex can be 3-deformed to a point (see [AC65]). Conversely, it is easy to see that every LOT complex can be transformed to a contractible complex by the addition of a 2-cell. This is why LOT complexes are considered test cases of the Whitehead conjecture. In [How85] he also proved the asphericity of LOTs of diameter ≤ 3 .

There has been remarkable progress in this problem in the last decades. Huck and Rosebrock [HR92, HR95] introduced two variations of Gersten's weight test [Ger87] to study asphericity of LOTs. In [Ros94] Rosebrock exhibited a class of LOTs which present hyperbolic groups different from the infinite cyclic group. This shows that the class of LOT groups is strictly larger than the class of knot groups. Huck and Rosebrock [HR01] proved the asphericity of prime injective LOTs and that every LOT can be reoriented to become aspherical. Later [HR07] they used a classical Lemma of Stallings about graphs on the 2-sphere to prove the asphericity of new classes of LOTs. Harlander and Rosebrock [HR17] recently proved the asphericity of all injective LOTs.

We refer the reader to [Ros07, Bog93, MR17] for more details and results about this problem.

Related to the study of asphericity of LOTs are the following group theoretic concepts.

Definition 1.2. A group G is called *indicible* if it has an infinite abelianization. Equivalently, G is indicible if there is a non-trivial map from G to the infinite cyclic group.

A group G is called *locally indicible* (LI) if every non-trivial finitely generated subgroup of G is indicible.

Remark 1.3. If a LOT group is locally indicible, then the LOT is aspherical. This is because LOT complexes have the homology of S^1 and locally indicible groups are conservative, therefore a universal cover of the LOT complex must have trivial second homology group (see Section 5 of [How85]).

It is not known whether all LOT groups are LI. In the present work we prove that a large class of LOTs have locally indicible LOT group. This class includes non-injective LOTs of any diameter and complexity. These results also provide a way of producing presentations of locally indicible groups.

An essential tool that we introduce in Section 2 is the concept of *good enumeration* for LOTs, which is, in some sense, a generalization of the concept of complexity introduced by S. Rosebrock [Ros10] and which is recalled below.

Definition 1.4. Given a LOG Γ we say that Γ has *complexity* N provided there is a subset S of the set of vertices of Γ consisting of N elements, such that the following inductive process generates all vertices of Γ and there is no such set consisting of $N - 1$ elements:

- The elements of S are generated.
- If v is incident to an edge such that the other incident vertex as well as the label vertex are generated, then v is generated.

For example, in the LOT of Figure 1, the set $S = \{a, b, c\}$ does not generate Γ , but the set $\{d, e\}$ does. In this case the complexity is 2.

Rosebrock proved that LOTs of complexity 2 are aspherical and that, in the case of LOIs (that is, LOTs for which the underlying graph is an interval), the complexity is bounded by $(n + 1)/2$, where n is the number of vertices. Later, Christmann and De Wolff [CdW14] proved that this upper bound for complexity extends to all LOTs. They also proved that if a LOT has maximal complexity, then it is a union of sub-LOTs of complexity 2 and therefore aspherical.

We recall finally a result of Howie, which we will apply in the next section.

Theorem 1.5. (Howie [How82]) *Let A and B be locally indicable groups, and let G be the quotient of $A * B$ by the normal closure of a cyclically reduced word r of length at least 2. Then the following are equivalent.*

- G is locally indicable.
- G is torsion free.
- r is not a proper power in $A * B$.

2 Main results

The following Lemma is a generalization of a Lemma of Rosebrock (see [Ros10], Lemma 2.2). In our notation G' means the abelianization $G/[G, G]$ of the group G .

Lemma 2.1. *Let $\langle X \mid R \rangle$ be a presentation of a group G such that $G' \simeq \mathbb{Z}$. Let H be the group presented by $\langle X, b \mid R, w^n \rangle$, where b is a new generator and w is a word in $X \cup \{b\}$. If $H' \simeq \mathbb{Z}$, then $n = 1$.*

Proof. Assume $n > 1$. Let p be the quotient map $G * \mathbb{Z} \rightarrow H$. There is a commutative diagram

$$\begin{array}{ccc} G * \mathbb{Z} & \xrightarrow{p} & H \\ \downarrow \pi & & \downarrow \pi \\ G' \oplus \mathbb{Z} & \xrightarrow{\bar{p}} & H' \end{array}$$

where the vertical arrows are the abelianization maps and $\bar{p}$ is the natural induced map. Then $\bar{p}$ is surjective and $\ker \bar{p} = \pi(\ker p)$ is generated by $\pi(w^n)$.

If $w = x_1 b^{i_1} x_2 b^{i_2} \cdots x_k b^{i_k}$, then $\pi(w) = \pi(x_1 \cdots x_k) + \pi(b^{i_1 + \cdots + i_k})$. Since $p(w^n) = 1$ then $n\pi(p(w)) = \pi(p(w^n)) = 0$, which implies $\pi(p(w)) = 0$, that is, $\bar{p}(\pi(w)) = 0$.

Therefore $\pi(w) \in \ker(\bar{p})$,

$$\pi(x_1 \cdots x_k) + \pi(b^{i_1 + \cdots + i_k}) \in \langle n(\pi(x_1 \cdots x_k) + \pi(b^{i_1 + \cdots + i_k})) \rangle$$

and then

$$\pi(x_1 \cdots x_k) \in \langle n\pi(x_1 \cdots x_k) \rangle \text{ and } \pi(b^{i_1 + \cdots + i_k}) \in \langle n\pi(b^{i_1 + \cdots + i_k}) \rangle$$

which is a contradiction unless $\pi(x_1 \cdots x_k) = 0$ and $\pi(b^{i_1 + \cdots + i_k}) = 0$. But in this case $x_1 \cdots x_k \in [G, G]$ and $i_1 + \cdots + i_k = 0$, thus $w \simeq x_1 \cdots x_k \bmod [G, b]$, where $[G, b]$ denotes the commutator of the subgroups $p(G)$ and $p(\langle b \rangle)$ of H . Therefore $w \in [G, G][G, b]$ and this implies that $H' = (G * b/N(w^n))' = (G * b)' \simeq \mathbb{Z} \oplus \mathbb{Z}$, which is also a contradiction. $\square$

We now introduce the concept of a *good enumeration* for a LOT, which is a key tool in this paper and appears in the hypotheses of the main theorem. We also present some properties about LOTs which admit a good enumeration, some of which will be needed for our main result. These results also yield a new tool for studying asphericity and local indicability.

The concept of good enumeration is, in some sense, related to the concept of complexity, as it is shown below (see Propositions 2.7 and 2.8). In fact, the proof of Theorem 2.9 is based in the ideas of [Ros10].

Definition 2.2. Let Γ be a LOT. A *good enumeration* of Γ is an enumeration of the edges of Γ in consecutive rows $F_1, F_2, \dots, F_r$,

$$\begin{aligned} F_1 : & x_{11}, x_{12}, \dots, x_{1q_1} \\ F_2 : & x_{21}, x_{22}, \dots, x_{2q_2} \\ & \dots \\ F_r : & x_{r1}, x_{r2}, \dots, x_{rq_r}, \end{aligned}$$

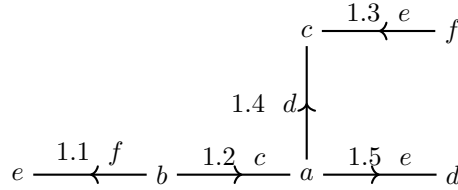
such that the following conditions are satisfied.

1. A new edge can be enumerated x_{ij} if it has one endpoint v_{ij} for which all the remaining incident edges, and the edges labeled by v_{ij} were previously enumerated as x_{kl} with $k < i$ or $k = i, l < j$, or if the edge x_{ij} starts a new row (i.e. if $j = 1$).
2. For every row F_s , $1 \leq s \leq r$ there exists a vertex w_s different from all the v_{ij} such that all the edges incident to w_s and all the edges labeled by w_s are enumerated x_{kl} with $k \leq s$.

We will say that a vertex v of Γ is *covered* if all the edges incident to v and all the edges labeled by v are enumerated. A subset of Γ will be said to be *covered* if all its vertices are covered. Thus condition 2 can be restated as follows:

For every $1 \leq s \leq r$ there exists a vertex w_s different from all the v_{ij} that is covered by the first s rows $F_1, \dots, F_s$.

For example, the LOT of Figure 1 admits the good enumeration in one row presented below,



taking $v_{1.2} = b, v_{1.3} = f, v_{1.4} = c, v_{1.5} = a, w_1 = d$.

In Figure 2 we present a LOT which admits a good enumeration in three rows. We omit the orientation in the edges, since it is irrelevant for the study of good enumeration. In fact, the results in the present work are independent of the orientation. The enumeration on the vertices is: $v_{1.2} = a, w_1 = i, v_{2.2} = m, v_{2.3} = l, v_{2.4} = k, v_{2.5} = c, v_{2.6} = d, v_{2.7} = b, v_{2.8} = h, w_2 = j, v_{3.2} = e, w_3 = g$.

Given a LOT Γ with a good enumeration, we will call Γ_1 the subset of vertices and edges of Γ obtained when omitting the first row, that is, omitting the edges $x_{1j} : 1 \leq j \leq q_1$ and the vertices $v_{1j} : 2 \leq j \leq q_1$ and w_1 . Similarly, we will call Γ_s the subset of Γ obtained when omitting all rows F_i with $i \leq s$.

Proposition 2.3. *Let Γ be a LOT which admits a good enumeration $\{x_{ij} : 1 \leq j \leq q_i, 1 \leq i \leq r\}$. Then Γ_s is a sub-LOT of Γ for all $1 \leq s \leq r$.*

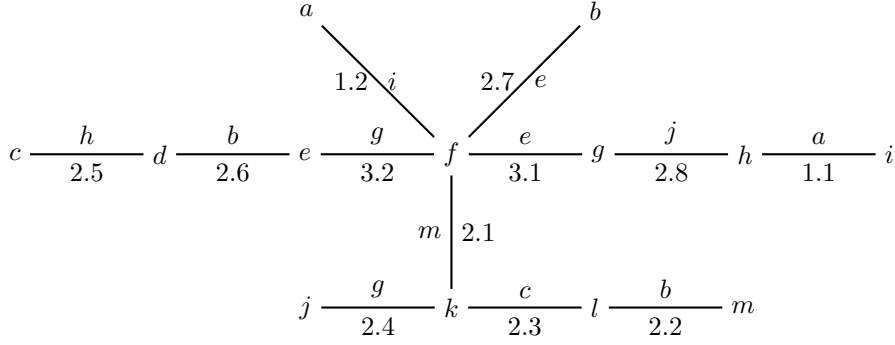


Figure 2: A good enumeration in three rows.

Proof. To see that Γ_1 is a subgraph, note that when a vertex v_{1j} of the first row is used to enumerate a new edge x_{1j} , it gets covered. Therefore v_{1j} and w_1 are all covered. This implies that the edges of Γ_1 have their ends in Γ_1 . Γ_1 is a tree because it cannot have cycles, and it has the same Euler characteristic as Γ . Finally, since v_{1j} and w_1 are all covered, the label vertices of Γ_1 are all in Γ_1 . Therefore Γ_1 is a sub-LOT. The proof is similar for Γ_s when $s > 1$. $\square$

Lemma 2.4. *The condition of the existence of w_s in the definition of a good enumeration is equivalent to the following:*

There is a component of $\Gamma - \{x_{s1}\}$ which is covered by the first s rows.

Proof. Assume that there is such a vertex w_s . Note that $\Gamma - \{x_{s1}\}$ has two components. Let us denote by $\Gamma_s = \Gamma - F_1, F_2, \dots, F_s$ the sub-LOT of Γ obtained when omitting the edges and vertices corresponding to the rows $F_1, F_2, \dots, F_s$. Then Γ_s is included in $\Gamma - \{x_{s1}\}$. Since Γ_s is connected, it must be included in one of the components. It is easy to check that the other component is covered.

Conversely, let C_s be the covered component. Then $\Gamma_{s-1} \cap C_s$ must be connected, so it must have one more vertex than its number of edges. The edges in $\Gamma_{s-1} \cap C_s$ are all of the form x_{sj} with $j > 1$, and each of these must have an associated vertex v_{sj} in $\Gamma_{s-1} \cap C_s$. Therefore there is one extra vertex in $\Gamma_{s-1} \cap C_s$. This vertex satisfies the conditions of w_s . $\square$

Proposition 2.5. *Let Γ be a LOT which admits a good enumeration, and let $\Gamma' \subseteq \Gamma$ be a sub-LOT. Then Γ' admits a good enumeration.*

Proof. Let $\{x_{ij} : 1 \leq j \leq q_i, 1 \leq i \leq r\}$ be a good enumeration of Γ . We may restrict this enumeration to Γ' , preserving rows and the order in each row. Eventually some row may not appear in Γ' . Let us call $\{x'_{kl} : 1 \leq l \leq q'_l, 1 \leq k \leq r'\}$ the restricted enumeration for Γ' . We must see that it satisfies conditions 1 and 2 of the definition.

Note that for each new edge x'_{kl} to be enumerated in Γ' , if $x'_{kl} = x_{ij}$, then the same vertex v_{ij} can be used as v'_{kl} . It remains for us to check the existence of the vertices w'_s . Let $1 \leq s \leq r'$, and assume that the row s of Γ' corresponds to the row t of Γ .

Since $\Gamma' \cap \Gamma_\alpha$ must be connected for every α , we know that the number of vertices and edges of each row that are missing in Γ' are equal. Besides, every edge $x'_{sl} = x_{tj}$ with $j > 1$ in $F_s = F_t \cap \Gamma'$ has its associated vertex $v'_{sl} = v_{tj}$ in Γ' . Therefore, if x_{t1} is in Γ' there must be one extra vertex of F_t in Γ' . It may be w_t or some v_{tj} . In both cases we can take this extra vertex to be w'_s .

Now, if $x_{t1} \notin \Gamma'$, then $x'_{s1} = x_{tj}$ with $j > 1$. In this case we may take $w'_s = v_{tj}$. $\square$

Corollary 2.6. *Let Γ be a LOT. If a row of edges $F_1 = \{x_{1j} : 1 \leq j \leq q_1\}$ of Γ can be enumerated satisfying condition 1 of the hypotheses of a good enumeration (see Definition 2.2) and if there is a vertex w_1 satisfying condition 2 of the definition, then Γ admits a good enumeration if and only if Γ_1 admits a good enumeration.*

This implies that, in the process of finding a good enumeration for a given LOT Γ , if one can enumerate a first row F_1 with its corresponding final vertex w_1 , this row can be omitted, and the problem reduces to finding a good enumeration for the remaining sub-LOT.

The following results relate good enumeration with the notion of complexity.

Proposition 2.7. *A LOT Γ has complexity 2 if and only if Γ admits a good enumeration in one row.*

Proof. Assume there are two vertices a, b generating Γ . Let us call $t_1 = a, t_2 = b, t_3, \dots, t_n$ the vertices of Γ in the order of the generating process, and e_k the edge which is used for generating t_k ($3 \leq k \leq n$). There is exactly one edge in Γ which is not used in this process, and we shall call it $\hat{e}$. We will prove that the enumeration given by $x_1 = \hat{e}, x_j = e_{n+2-j}, v_j = t_{n+2-j}$ ($2 \leq j \leq n-1$) is a good enumeration for Γ . To see this we must check that every v_j satisfies that, except for the edge x_j , all other incident edges and all other edges with label v_j are enumerated x_i with $i < j$.

Let us set $k = n + 2 - j$. Then $v_j = t_k$ and $x_j = e_k$, and the edge e_k is used, in the generating process, to generate the vertex t_k . Therefore all other edges involving this vertex (as an endpoint or label) must be e_l with $l > k$ or $\hat{e}$. This means that they are enumerated as x_i with $i < j$.

The converse implication can be proved in a similar way. $\square$

Proposition 2.8. *Let Γ be a LOT. If Γ admits a good enumeration in N rows, then Γ has complexity at most $N + 1$.*

Proof. Using the notation of Definition 2.2, we consider the set $S = \{w_0, w_1, \dots, w_N\}$ where w_0 denotes the only vertex of Γ which is not involved in the enumeration. Let us see that this is a generating set. We start with the last vertex v_{N,q_N} of the last row. The edge x_{N,q_N} was enumerated using this endpoint, therefore the label vertex cannot be v_{N,q_N} and neither the other endpoint of this edge nor its label vertex can be vertices enumerated before this edge. Thus they cannot be enumerated at all, and must be vertices in S . Then the vertex v_{N,q_N} is generated. Following the inverse order of the enumeration, every vertex v_{ij} is generated through its associated edge x_{ij} , using the fact that its label and other endpoint were generated before. $\square$

We now prove the main result of this article.

Theorem 2.9. *Let Γ be a LOT which admits a good enumeration. Then the LOT group $G(\Gamma)$ is locally indicable.*

Proof. We proceed by induction on N , the number of rows of a good enumeration for Γ .

Case $N = 1$. Let $x_1, \dots, x_q$ be a good enumeration of the edges of Γ , with associated vertices $v_2, \dots, v_q$ and w_1 . Let w_0 be the only vertex which is not involved in the enumeration. Let us consider the LOT presentation with a convenient order: $P(\Gamma) = \langle w_0, w_1, v_2, \dots, v_q \mid r_1, \dots, r_q \rangle$, where r_j denotes the relator arising from the edge x_j . Depending on the orientation of the edge x_q , the relator r_q is either $\lambda v_q \lambda^{-1} u^{-1}$ or $\lambda u \lambda^{-1} v_q^{-1}$ for some vertices λ, u . We use this relator to express v_q as a word in w_0 and w_1 (which is possible because of the good enumeration). Then we can replace all the occurrences of v_q in other relators with this word, and then omit the generator v_q and the relator r_q . In this way we obtain a new presentation $P^{(1)} = \langle w_0, w_1, v_2, \dots, v_{q-1} \mid r_1^{(1)}, \dots, r_{q-1}^{(1)} \rangle$, where $r_j^{(1)}$ denotes the relator obtained when replacing the occurrences of v_q for the corresponding word in w_0, w_1 . We proceed in a similar way with the generators $v_{q-1}, v_{q-2}, \dots, v_2$ and obtain $P^{(q-1)} = \langle w_0, w_1 \mid r_1^{(q-1)} \rangle$. This procedure corresponds to a sequence of Q^{**} operations, and

in particular $P^{(q-1)}$ is a presentation of $G(\Gamma)$. Note that $G(\Gamma)$ is LI because of Lemma 2.1 and Theorem 1.5.

For the inductive step, consider a good enumeration in N rows

$$\begin{aligned} & x_{11}, x_{12}, \dots, x_{1q_1} \\ & x_{21}, x_{22}, \dots, x_{2q_2} \\ & \dots \\ & x_{N1}, x_{N2}, \dots, x_{Nq_N}, \end{aligned}$$

with corresponding vertices $\{v_{ij} : 1 \leq i \leq N, 2 \leq j \leq q_i\}$ and $\{w_i : 1 \leq i \leq N\}$. Let w_0 be the only vertex of Γ not involved in the enumeration. Then

$$P(\Gamma) = \langle w_0, w_1, \dots, w_N, v_{12}, \dots, v_{1q_1}, \dots, v_{N2}, \dots, v_{Nq_N} \mid r_{11}, \dots, r_{1q_1}, \dots, r_{N1}, \dots, r_{Nq_N} \rangle.$$

We proceed in a similar way as in the case $N = 1$, using the relators of the first row, starting from r_{1q_1} , to write the generators v_{1j} as words in the generators w_i , replacing them in all other relators, and omitting these generators and relators from the presentation. During this process, the relators associated to succeeding rows are not modified. With these Q^{**} operations we obtain a new presentation

$$\tilde{P} = \langle w_0, w_1, \dots, w_N, v_{22}, \dots, v_{2q_2}, \dots, v_{N2}, \dots, v_{Nq_N} \mid \tilde{r}_{11}, r_{21}, \dots, r_{2q_2}, \dots, r_{N1}, \dots, r_{Nq_N} \rangle.$$

On the other hand, we consider the sub-LOT Γ_1 of Γ , which admits a good enumeration in $N - 1$ rows, and for which the presentation is

$$P(\Gamma_1) = \langle w_0, w_2, \dots, w_N, v_{22}, \dots, v_{2q_2}, \dots, v_{N2}, \dots, v_{Nq_N} \mid r_{21}, \dots, r_{2q_2}, \dots, r_{N1}, \dots, r_{Nq_N} \rangle.$$

Therefore $\tilde{P}$ is obtained from $P(\Gamma_1)$ by adding the generator w_1 and the relator $\tilde{r}_{11}$.

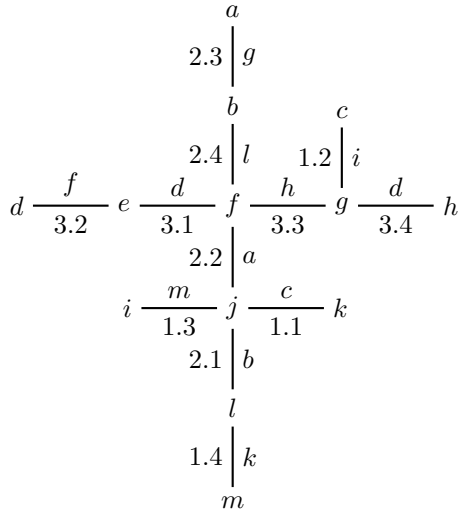
By induction, $G(\Gamma_1)$ is LI, and because of Lemma 2.1 and Theorem 1.5, $G(\Gamma)$ is LI. □

As an immediate consequence we obtain the following.

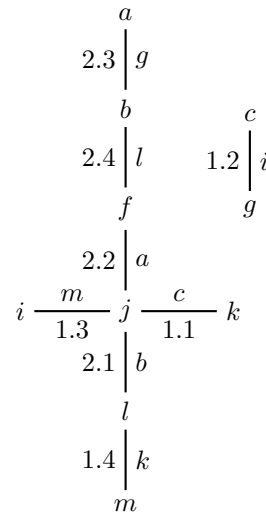
Corollary 2.10. *Let Γ be a LOT which admits a good enumeration. Then Γ is aspherical.*

We illustrate with an example that this result can be applied to a wide class of LOTs, including LOTs of large diameter, large number of vertices, and high complexity, and as a consequence it yields a way of producing substantial presentations of locally indicable groups.

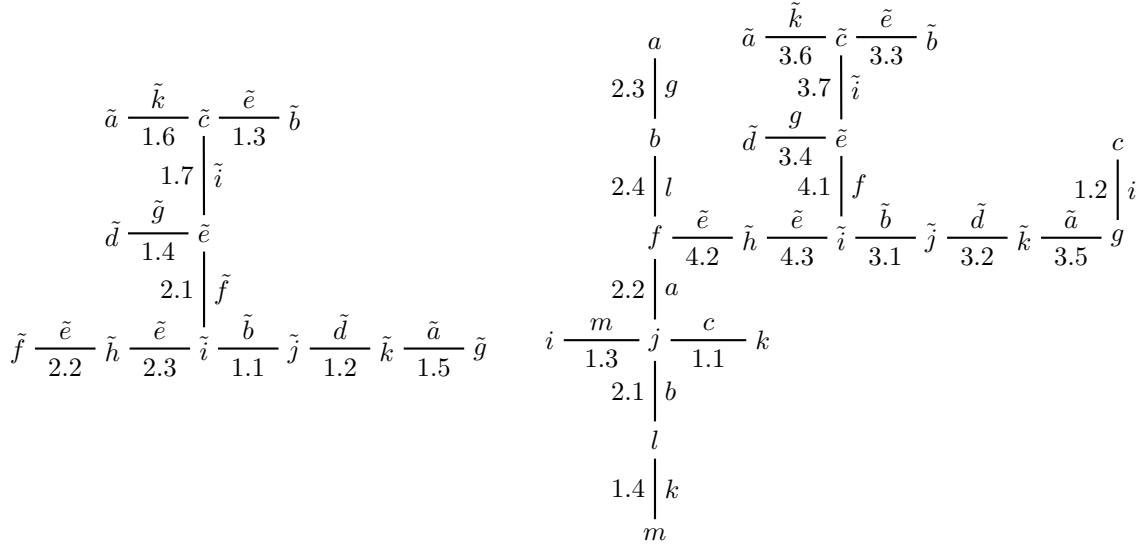
The LOT of Figure 3a admits a good enumeration in three rows (with any orientation of the edges). If we remove all the edges corresponding to the last row, and the vertices which are incident only to such edges, as is shown in Figure 3b we obtain a sub-LOG (note that in general the result of doing this can be non-connected). Combining this sub-LOG with any LOT which admits a good enumeration and such that the resulting graph is connected yields a new LOT with a good enumeration. For example, taking the LOT of figure 3c and identifying $f \sim \tilde{f}, g \sim \tilde{g}$, this procedure yields the LOT shown in Figure 3d.



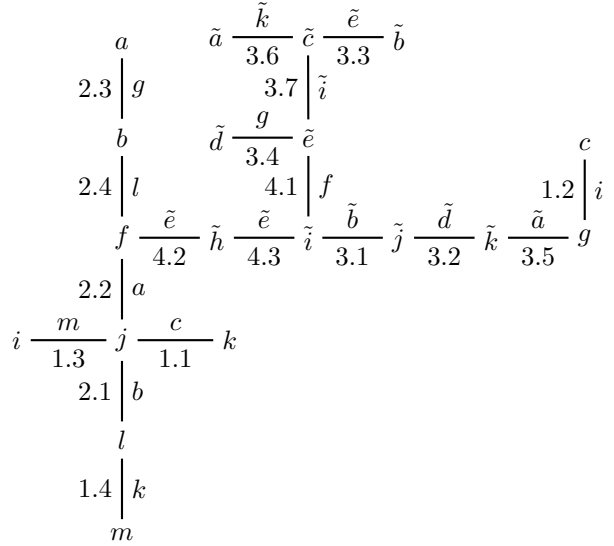
(a) A LOT with a good enumeration in 3 rows



(b) A sub-LOG



(c) Another LOT with a good enumeration



(d) A new LOT with a good enumeration

Figure 3: Creating a new LOT with a good enumeration.

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